

## 1 Basic Quantities

- **Accumulation Function:**  $a(t)$ , where  $\$P$  grows to  $\$P \cdot a(t)$  at time  $t$  and  $a(0) = 1$
- **Inverse Accumulation Function:**  $v(t) = [a(t)]^{-1}$
- **Interest Force:**  $\delta_t = \frac{d}{dt} \ln a(t)$ , where  $a(t) = \exp(\int_0^t \delta_x dx)$ ;  $\delta = \ln(1+i)$  and  $a(t) = \exp(\delta t)$  for compound interest
- **Annual Effective Interest Rate:**  $i_n = \frac{a(n)-a(n-1)}{a(n-1)}$  for year  $n$ , and  $a(n) = (1+i)^n$  for same  $i$  over years
- **Annual Effective Discount Rate:**  $d_n = \frac{a(n)-a(n-1)}{a(n)}$  for year  $n$ , and  $a(n) = (1-d)^{-n}$  for same  $d$  over years
- **Interest-Discount Transformation:**  $d = \frac{i}{1+i}, i = \frac{d}{1-d}$
- **Nominal Interest Rate Compounded  $m$  Times Annually:**  $i^{(m)}$ , where  $(1 + \frac{i^{(m)}}{m})^m = (1+i)$
- **Nominal Discount Rate Compounded  $m$  Times Annually:**  $d^{(m)}$ , where  $(1 - \frac{d^{(m)}}{m})^{-m} = (1+i)$
- **Growth Factor:**  $j$ , where  $\$R$  grows to  $\$R(1+j)$  in the next period
- **Inverse Growth Factor:**  $v = (1+j)^{-1}$ , where  $\$R$  drops to  $\$Rv$  in the previous period
- **Discount Factor:**  $d = 1 - v = 1 - (1+j)^{-1}$ , where

$$v = 1 - d, \quad \frac{d}{v} = \frac{1-v}{v} = v^{-1} - 1 = j, \quad (1-d)^{-1} = (1+j), \quad \frac{1}{1-v} = \frac{1+j}{j}$$

## 2 Annuity

- PV of annuity with  $n$  periods of immediate/due payment of  $\$R$ , at interest rate  $j$  per period:

$$R \cdot a_{\overline{n}|j} = R(v + \cdots + v^n) = R \cdot \frac{1-v^n}{j}, \quad v = (1+j)^{-1} \quad (\text{immediate})$$

$$R \cdot \ddot{a}_{\overline{n}|j} = R(1 + \cdots + v^{n-1}) = R \cdot \frac{1-v^n}{d}, \quad d = 1 - v \quad (\text{due})$$

- FV of annuity with  $n$  periods of immediate/due payment of  $\$R$ , at interest rate  $j$  per period:

$$R \cdot s_{\overline{n}|j} = R(1 + \cdots + a^{n-1}) = R \cdot \frac{a^n - 1}{j}, \quad a = (1+j) = v^{-1} \quad (\text{immediate})$$

$$R \cdot \ddot{s}_{\overline{n}|j} = R(a + \cdots + a^n) = R \cdot \frac{a^n - 1}{d}, \quad a = (1+j) = v^{-1} \quad (\text{due})$$

- PV of annuity with  $n$  years,  $m$  equal payments per year, annual payment  $\$R$ , and annual interest rate  $i$ :

$$\frac{R}{m} \cdot a_{\overline{n}|j}^{(m)} = \frac{R}{m} \cdot \frac{1-v^n}{i^{(m)}} \quad (\text{immediate})$$

$$\frac{R}{m} \cdot \ddot{a}_{\overline{n}|j}^{(m)} = \frac{R}{m} \cdot \frac{1-v^n}{d^{(m)}} \quad (\text{due})$$

- FV of annuity with  $n$  years,  $m$  equal payments per year, annual payment  $\$R$ , and annual interest rate  $i$ :

$$\frac{R}{m} \cdot s_{\overline{n}|j}^{(m)} = \frac{R}{m} \cdot \frac{(1+j)^n - 1}{i^{(m)}} \quad (\text{immediate})$$

$$\frac{R}{m} \cdot \ddot{s}_{\overline{n}|j}^{(m)} = \frac{R}{m} \cdot \frac{(1+j)^n - 1}{d^{(m)}} \quad (\text{due})$$

- PV of perpetuity with infinite periods of immediate/due payment of  $\$R$ , at interest rate  $j$  per period:

$$R \cdot a_{\infty|j} = R \cdot \frac{1}{j} \quad \leftrightarrow \quad R \cdot \ddot{a}_{\infty|j} = R \cdot \frac{1}{d} \quad (\text{Perpetuity})$$

### 3 Annuity with Varying Payments

Assume principle  $\$P$ , increment  $\$Q$ , term  $n$ , period interest  $j$ , and  $v = (1 + j)^{-1}$ ,  $d = 1 - v$ :

- PV immediate/due:

$$Pa_{\overline{n}|j} + Q \left( \frac{a_{\overline{n}|j} - nv^n}{j} \right) \leftrightarrow P\ddot{a}_{\overline{n}|j} + Q \left( \frac{a_{\overline{n}|j} - nv^n}{d} \right)$$

- FV immediate/due:

$$Ps_{\overline{n}|j} + Q \left( \frac{s_{\overline{n}|j} - n}{j} \right) \leftrightarrow P\ddot{s}_{\overline{n}|j} + Q \left( \frac{s_{\overline{n}|j} - n}{d} \right)$$

If  $\boxed{P = Q = 1}$ :

- PV immediate/due:

$$(Ia)_{\overline{n}|j} = \frac{\ddot{a}_{\overline{n}|j} - nv^n}{j} \leftrightarrow (I\ddot{a})_{\overline{n}|j} = \frac{\ddot{a}_{\overline{n}|j} - nv^n}{d}$$

$$(Ia)_{\infty|j} = \frac{1}{j} + \frac{1}{j^2} \leftrightarrow (I\ddot{a})_{\infty|j} = \frac{1}{d^2}$$

- FV immediate/due:

$$(Is)_{\overline{n}|j} = \frac{\ddot{s}_{\overline{n}|j} - n}{j} \leftrightarrow (I\ddot{s})_{\overline{n}|j} = \frac{\ddot{s}_{\overline{n}|j} - n}{d}$$

If  $\boxed{P = 1, Q = -1}$ :

- PV immediate/due:

$$(Da)_{\overline{n}|j} = \frac{n - a_{\overline{n}|j}}{j} \leftrightarrow (D\ddot{a})_{\overline{n}|j} = \frac{n - a_{\overline{n}|j}}{d}$$

- FV immediate/due:

$$(Ds)_{\overline{n}|j} = \frac{n(1 + j)^n - s_{\overline{n}|j}}{j} \leftrightarrow (D\ddot{s})_{\overline{n}|j} = \frac{n(1 + j)^n - s_{\overline{n}|j}}{d}$$

**Deferred Annuity:** If deferred  $k$  periods, PV immediate/due:

$${}_k|a_{\overline{n}|j} = v^k a_{\overline{n}|j} \leftrightarrow {}_k|\ddot{a}_{\overline{n}|j} = v^k \ddot{a}_{\overline{n}|j}$$

**Continuous Annuity:**

- PV with rate of payment  $f(t)$  (pay  $f(t)dt$  during  $[t, t + dt]$ ):

$$PV = \int_0^n v(t)f(t)dt$$

- FV with rate of payment  $f(t)$ :

$$FV = PV \cdot a(n) = a(n) \int_0^n v(t)f(t)dt$$

- PV/FV for level  $f(t) = 1$ :

$$\bar{a}_{\overline{n}|j} = \frac{1 - v^n}{\delta} \leftrightarrow \bar{s}_{\overline{n}|j} = \frac{(1 + j)^n - 1}{\delta}$$

## 4 Amortization

Assume principle  $B_0$ , immediate-payment  $\$R$  (equal for all terms), term  $n$ , interest rate  $j$  per term. At the end of  $k$ -th period,  $k = 1, \dots, n$ ,

- **(Repaid) Interest:**  $I_k = B_{k-1} \cdot j$
- **(Repaid) Principle:**  $P_k = R - I_k$
- **Outstanding Balance:**  $B_k = B_{k-1} - P_k$
- **Header of Amortization Table:** |  $k$  |  $B_k$  |  $R$  |  $I_{k+1}$  |  $P_{k+1}$  |  $B_{k+1}$  |
- **Useful Formulas:**

$$B_k = \underbrace{B_0(1+j)^k - Rs_{\overline{k}|j}}_{\text{Retrospective}} = \underbrace{Ra_{\overline{n-k}|j}}_{\text{Prospective}}, \quad I_k = R(1 - v^{n-k+1}), \quad P_k = R - I_k = Rv^{n-k+1}$$

$$\frac{P_{k+1}}{P_k} = (1+j) \implies \frac{P_{k+n}}{P_k} = (1+j)^n$$

$$\text{Total Payment} = \sum_{k=1}^n P_k = nR, \quad \text{Total Interest} = \sum_{k=1}^n I_k = nR - B_0$$

- **Sinking Fund:** Assume principle  $\$P$ , term  $n$ , interest rate  $i$ , sinking fund interest rate  $j$ , sinking fund payment  $\$D$ . Target: Pay out interest  $\$Pi$  yielded during each period and accumulate regular payments of  $\$D$  to repay principle  $P$  at the end. Regular payment  $R = Pi + D = Pi + P/a_{\overline{n}|j} = P(i + 1/a_{\overline{n}|j})$ .

## 5 Bond

Assume Face (Par) Value  $F$ , term  $n$ , coupon rate (promised interest rate)  $\alpha$  compounded  $m$  times per year, yield rate (real interest rate)  $j$ , redemption value  $C$ . We say redemption at  $100 \cdot C/F$  (at par when  $C = F$ ).

- **Coupon (Paid per Period):**  $Fr = F \frac{\alpha}{m}$
- **Bond Price (Present Value):**  $P = C(1+j)^{-n} + (Fr)a_{\overline{n}|j} = C + (Fr - Cj)a_{\overline{n}|j}$
- Bond is purchased at **discount** if  $P < C \iff (Fr - Cj) < 0$ ; at **premium** if  $P > C \iff (Fr - Cj) > 0$ ; at **par** if  $P = C \iff Fr = Cj$ .
- **Book Value:**  $B_k = Fra_{\overline{n-k}|j} + C(1+j)^{n-k}$ , the remaining payments at time after  $k$ th payment.
- **Cleaned Dirty Price:**  $B_{t+k} = \underbrace{B_t(1+j)^k}_{\text{dirty price at } t+k} - \underbrace{kFr}_{\text{accrued interest } I}$ , where  $0 \leq k < 1$  and  $t+k$  denotes the partial period before the next coupon period

## 6 Rate of Return

Express time in years. Suppose a fund has value  $F_0$  at time 0,  $F_T$  at time  $T$  (the end), and  $n$  contributions  $C_{t_1}, \dots, C_{t_n}$  deposited at  $t_1, \dots, t_n$  respectively. Goal is to find the annual yield rate  $i$ .

- **Time- $T$  Value Equation:** Looking at investment from time 0 to time  $T$  where all cash flows  $C_{t_j}$  occur at time  $t_j \in [0, T]$  with initial fund  $F_0$  at time 0,

$$F_0(1+i)^T + \sum_{j=1}^n C_{t_j}(1+i)^{T-t_j} = F_T \text{ (typically choose } T=1\text{)}$$

The Time-0 Value Equation is thus

$$F_0 + \sum_{j=1}^n C_{t_j}(1+i)^{-t_j} = F_T(1+i)^{-T}$$

- **Dollar-Weighted Yield Rate (DWYR):** Assuming all cash flows occur in one year and approximating yield rate using simple interest  $(1+i)^{t_j} \approx (1+it_j)$ ,

$$i \approx \frac{I}{F_0 + \sum_{j=1}^n C_{t_j}(1-t_j)}$$

where  $I = F_T - F_0 - \sum_{j=1}^n C_{t_j}$  is the interest earned during the year. If have  $k \geq 2$  years, then  $t_j = \frac{\# \text{ years}}{k}$  which is the portion of the  $k$  years when the cash flow occurs, and  $i$  is the interest rate for  $k$  years (the real annual rate is thus  $(1+i^*)^k = (1+i) \implies i^* = (1+i)^{1/k} - 1$ ). If using *basic midpoint approximation* where all contributions are made at time  $\frac{1}{2}$ ,

$$i \approx \frac{2I}{F_0 + F_1 - I}$$

- **Time-Weighted Yield Rate (TWYR):**

$$(1+i)^T = (1+i_1)(1+i_2) \cdots (1+i_n) \\ = \left( \frac{F_{t_1}^-}{F_0} \right) \left( \frac{F_{t_2}^-}{F_{t_1}^+} \right) \cdots \left( \frac{F_T}{F_{t_n}^+} \right)$$

where  $(1+i_k)$  is the value right before payment  $k$  divided by right after payment  $k-1$  and

$$t_0 = 0, t_{n+1} = T \\ F_{t_k}^+ = F_{t_k}^- + C_{t_k}$$

- DWYR accounts for the times that cash flows occur, e.g. sell high and buy low can be reflected. YWYR ignores the timing of cash flows and only focuses on the investment's performance itself.
- **Minimum Yield Rate (Wanted by bond seller):** if  $t$  is the term of the bond, i.e. the time when bond's remaining values are paid out, the yield  $y_t$  varies as  $t$  changes:
  - at discount ( $P < C$ ): seller chooses maturity  $t$  as large as possible for minimum yield (later the smaller)
  - at premium ( $P > C$ ): seller chooses maturity  $t$  as small as possible for minimum yield (earlier the smaller)
  - at par (zero-coupon bond):  $t$  is indifferent
- **Spot Rates:**  $r_t$ , the *annual* effective interest rate for one single payment at time  $t$  invested at time 0 (no other payments between  $[0, t]$ , i.e. zero-coupon bond. In contrast, **Yield Rate**  $y_t$  is the one with payments in between). E.g. bonds:  $y_t = r_t$  when coupon is 0, otherwise

$$P = C(1+y_t)^{-t} + Fra_{\overline{t}|y_t} \quad (\text{in context of } y_t)$$

$$= Fr(1+r_1)^{-1} + Fr(1+r_2)^{-2} + \cdots + (C+Fr)(1+r_t)^{-t} \quad (\text{in context of } r_t)$$

where each coupon payment  $Fr$  at time  $k$  can be treated as a single payment of length  $k$  (thus has spot rate  $r_k$ ).

- **Forward Rate:** annual effective interest rate over period  $[s, t]$  as implied by  $r_s, r_t$  (which highly relates to  $r_s, r_t$ ):

$$(1+f_{[s,t]})^{t-s} = \frac{(1+r_t)^t}{(1+r_s)^s}$$

## 7 Asset-Liability Management

- **Price Curve:** Plot of price  $P(i)$  versus  $i$  where  $P(i)$  is the PV of cash flows  $C_{t_k}$  at  $t_k$  for  $k = 1, 2, \dots$  with annual effective interest rate  $i$ ,

$$P(i) = \sum_{k>0} C_{t_k} v_i^{t_k}$$

which is decreasing over  $i$ .

- **Duration** is the sensitivity of price due to change in interest rate  $i$  (could be in any shape,  $i, \delta, i^{(m)}$ , etc.). Small duration  $\implies$  price is stable when interest rate changes.
- **Macaulay Duration:**  $D(i, \infty) = -\frac{P'_\delta(i)}{P(i)}$  ( $> 0$  since  $P'_\delta(i) < 0$ ), the percentage change in  $P$  regarding  $\boxed{\delta = \ln(1+i)}$ ,

$$D(i, \infty) = -\frac{1}{P(i)} \frac{dP}{d\delta} = \frac{\sum_{k>0} t_k C_{t_k} v_i^{t_k}}{P(i)} = \frac{\sum_{k>0} t_k \cdot \text{PV of cash flow at } t_k}{\text{PV of all cash flows (Price)}}$$

where  $i^{(\infty)} = \delta = \ln(1+i)$  and  $\frac{dP}{di} = \frac{dP}{d\delta} \frac{d\delta}{di} = \frac{dP}{d\ln(1+i)} \frac{d\ln(1+i)}{di}$ . Proof uses  $v_i^t = e^{-\delta t}$ . Note it can be rearranged to  $\sum_{k>0} \left( \frac{C_{t_k} v_i^{t_k}}{P(i)} \right) t_k$  which is the sum of payment times weighted by cash flow importance (the proportion of this cash flow's PV in total PV).

- **Modified Duration:**  $D(i, m) = -\frac{P'_{i^{(m)}}(i)}{P(i)}$  ( $> 0$  since  $P'_{i^{(m)}}(i) < 0$ ), the percentage change in  $P$  regarding  $\boxed{i^{(m)}}$ , which is  $D(i, \infty)$  adjusted by chain rule factor  $\frac{d\delta}{di^{(m)}} = v_i^{1/m} = 1 + \frac{i^{(m)}}{m}$ ,

$$\begin{aligned} D(i, m) &= -\frac{1}{P(i)} \frac{dP}{di^{(m)}} = -\underbrace{\frac{1}{P(i)} \frac{dP}{d\delta}}_{D(i, \infty)} \frac{d\delta}{di^{(m)}} \\ &= D(i, \infty) \cdot v_i^{1/m} = \frac{D(i, \infty)}{1 + \frac{i^{(m)}}{m}} \\ &= \frac{\sum_{k>0} t_k C_{t_k} v_i^{t_k+1/m}}{P(i)} \end{aligned}$$

Note:

- Macaulay duration is tied to  $\delta$ ; Modified duration is tied to  $i^{(m)}$
- Macaulay duration is not necessarily needed to obtain Modified duration, and we don't necessarily rely on the summation formula. Just start with the derivative  $P'_{i^{(m)}}(i)$  and multiply it by  $-\frac{1}{P(i)}$ , then substitute in interest rate  $i$  to get duration.

- **Examples:**

- $n$ -year zero-coupon bond with redemption  $C$ :  $D(i, \infty) = n$ , i.e. duration = maturity.
- $n$ -year bond with annual coupon  $Fr$  and redemption  $C$ :

$$D(i, \infty) = \frac{Fr(Ia)_{\overline{n}|j} + nCv_i^n}{Fra_{\overline{n}|j} + Cv_i^n}$$

- $n$ -year annual annuity with level payment  $R$ :

$$D(i, \infty) = \frac{(Ia)_{\overline{n}|j}}{a_{\overline{n}|j}}$$

- **Macaulay Convexity:**

$$C(i, \infty) = \frac{1}{P(i)} \frac{d^2 P}{d\delta^2} = \frac{\sum_{k>0} t_k^2 C_{t_k} v_i^{t_k}}{P(i)}$$

- **Modified Convexity:**

$$C(i, m) = \frac{1}{P(i)} \frac{d^2 P}{di^{(m)2}} = \frac{C(i, \infty) + D(i, \infty)/m}{(1+i)^{2/m}}$$

Note it can be written as

$$\frac{\sum_{k>0} t_k(t_k + 1)C_{t_k}(1+i)^{-t_k-2}}{P(i)}$$

i.e. involves the derivative regarding  $i$  (might vary depending on timing).

- Change in  $P(i)$  if  $i_{old}^{(m)}$  changes to  $i_{new}^{(m)} = i_{old}^{(m)} + \Delta$ :

$$P(i)|_{i_{new}^{(m)}} - P(i)|_{i_{old}^{(m)}} = P(i)|_{i_{old}^{(m)}} \left[ -\Delta \cdot D(i, m)|_{i_{old}^{(m)}} + \frac{\Delta^2 \cdot C(i, m)|_{i_{old}^{(m)}}}{2!} \right]$$

and  $P(i)|_{i_{new}^{(m)}} = P(i)|_{i_{old}^{(m)}} \left[ 1 - \Delta \cdot D(i, m)|_{i_{old}^{(m)}} + \frac{\Delta^2 \cdot C(i, m)|_{i_{old}^{(m)}}}{2!} \right]$ . The minus sign exists because duration  $D$  is proportional to the negative of first-order derivative of  $P$ . Use  $D(i, m), C(i, m)$  since the change is related to  $i^{(m)}$ .

- **Immunization:** To pay off a liability at time  $T$  or multiple liabilities at different times, we need to balance the price effects and reinvestment risks. If there's exact cash flow matching (options of bonds that can provide cash inflows at exactly the times of liabilities) then we can buy a portfolio of them, specifying the face value of each bond to be purchased. Let  $A_t$  denote total cash inflow (assets) at time  $t$ ,  $L_t$  denote total cash outflow (liabilities) at time  $t$ , i.e. the net cash flow at  $t$  is  $A_t - L_t$ . The surplus (at time 0) is defined as

$$\begin{aligned} S(i) &= \sum_{t \geq 0} (A_t - L_t)(1+i)^{-t} \\ &= \sum_{t \geq 0} A_t(1+i)^{-t} - \sum_{t \geq 0} L_t(1+i)^{-t} \\ &= P_A(i) - P_L(i) \end{aligned}$$

We want to make sure the asset-liability balance is stable, i.e.

1.  $S(i_0) = 0$  i.e.  $PV(\text{assets}) = PV(\text{liabilities})$
2.  $S'(i_0) = 0$  i.e. stationary point of a convex function
3.  $S''(i_0) \geq 0$  i.e. convex, so that  $i$ 's small fluctuations (both up and down) will always lead to a positive surplus

These conditions are equivalent to

1.  $P_A(i_0) = P_L(i_0)$   
i.e. assets and liabilities have equal present values
2.  $D_A(i_0, \infty) = D_L(i_0, \infty)$   
i.e. assets and liabilities have equal Macaulay durations so that price and reinvestment effects cancel out
3.  $C_A(i_0, \infty) \geq C_L(i_0, \infty)$   
i.e. assets' Macaulay duration is at least liabilities', which ensures assets' duration is at least liabilities'